

Kseniia Prokofeva, PhD

Behavioral and systems neuroscientist studying the neural mechanisms of other-oriented social behavior

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[Google Scholar](#)
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Academic Appointments

2023 – Present	Assistant Professor Institute of Science Tokyo , Japan School of Life Science and Technology, Laboratory of Kumi O. Kuroda, MD, PhD
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Education

2019 – 2023	Doctor of Philosophy in Medical Sciences University of Tsukuba , International Institute for Integrative Sleep Medicine, Japan Thesis: Regulation of wakefulness and sleep by neuronal circuits linking ventrolateral preoptic nucleus and lateral hypothalamic area Advisors: Takeshi Sakurai, MD, PhD, and Arisa Hirano, PhD
2013 – 2019	Specialist Diploma in Pediatrics Novosibirsk State Medical University , Faculty of Pediatrics, Russian Federation Qualification: Pediatrician

Research Funding, Fellowships & Honors

2025 – 2027	JSPS KAKENHI Grant-in-Aid for Early-Career Scientists, PI, Project No. 25K18581, Institute of Science Tokyo, Japan, ¥4.68M
2019 – 2023	Japanese Government (MEXT) Scholarship, University Recommendation Full doctoral scholarship, University of Tsukuba, Japan
2019	Diploma with Honors, Novosibirsk State Medical University, Russia

Peer-Reviewed Publications

* co-first authors; # corresponding authors

Prokofeva K*, Shibamiya M*, Kawata R, Yoshihara C, Kuroda KO#. (2026) Neural correlates of adversity-overcoming pup rescue behavior in female mice. *Scientific Reports*. 16, 11844. [doi: 10.1038/s41598-026-35639-7](https://doi.org/10.1038/s41598-026-35639-7)

Prokofeva K, Saito YC, Niwa Y, Mizuno S, Takahashi S, Hirano A#, Sakurai T#. (2023) Structure and Function of Neuronal Circuits Linking Ventrolateral Preoptic Nucleus and Lateral Hypothalamic Area. *The Journal of Neuroscience*. 43(22):4075-4092. [doi: 10.1523/JNEUROSCI.1913-22.2023](https://doi.org/10.1523/JNEUROSCI.1913-22.2023)

Current Research

2023 – Present	Institute of Science Tokyo , Japan - Investigate neural mechanisms of infant care and helping-like behavior under environmental adversity in female mice. - Co-developed and refined behavioral paradigms examining infant-directed care and cagemate-directed helping across parametrically varied environmental costs. - Coordinate projects integrating quantitative behavioral analysis with neural activity mapping, AAV-mediated cell ablation, chemogenetic and optogenetic manipulation, and fiber photometry.
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Research Supervision & Mentoring

2023 – Present	Provided day-to-day research supervision to four undergraduate students; three subsequently progressed to Master's study and continue under my guidance, while one is currently an undergraduate. ▪ Guide projects investigating the neural mechanisms of infant care, helping-like behavior, and sexual behavior. ▪ Provide guidance in experimental design and procedures, behavioral experimentation, quantitative data analysis, interpretation, and scientific presentation. ▪ Support preparation of undergraduate graduation research and Master's research presentations. ▪ One Master's student under my day-to-day guidance received a university-level poster award for research presentation (Excellent Poster Award at the FY2025 Interim Presentation Meeting for Master's Thesis Research Plans). ▪ Support students in developing increasing independence in experimental planning, data interpretation, and scientific presentation.
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Teaching Experience

Oct 2026 – Present	Instructor, Scientific discussions in English, Institute of Science Tokyo, Japan
2025 – Present	Instructor, Functional Life Science, Institute of Science Tokyo, Japan
2023 – Present	Instructor, LST Basic Laboratory and Exercise III, Institute of Science Tokyo, Japan
2021	Graduate Teaching Assistant, Summer Research Program 2021, University of Tsukuba, Japan

Methodological Expertise

▪ Behavioral neuroscience: design and quantitative analysis of mouse social behavior paradigms; cost/adversity manipulations; automated pose tracking with DeepLabCut. ▪ Systems neuroscience: fiber photometry; optogenetic and chemogenetic manipulations; stereotaxic viral approaches. ▪ Neuroanatomy and activity mapping: circuit tracing; immunohistochemistry; in situ hybridization; c-Fos mapping. ▪ Quantitative analysis: R; mixed-effects and longitudinal modeling; survival analysis; behavioral and neural time-series analysis.
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Conference Presentations

Oral Presentations

Prokofeva K, Shibamiya M, Kawata R, Yoshihara C, Kuroda KO. Neural correlates of adversity-overcoming pup rescue in female virgin mice. *The 39th Meeting of the Japanese Society for Behavioral Neuroendocrinology* (August 2025). Kyoto, Japan.

Prokofeva K, Shibamiya M, Kawata R, Yoshihara C, Kuroda KO. A scalable paradigm for investigating pup rescue behavior under adversity in female mice. *The 48th Annual Meeting of the Japan Neuroscience Society* (July 2025). Niigata, Japan.

Prokofeva K, Saito YC, Hirano A, Sakurai T. Delineation of Neural Circuits of Galaninergetic Neurons in the VLPO Implicated in Regulation of Sleep. *The 45th Annual Meeting of the Japan Neuroscience Society* (June 2022). Ginowan, Japan.

Oral Presentations

Prokofeva K, Saito YC, Hirano A, Sakurai T. Mapping of presynaptic partners of the sleep-implicated preoptic – lateral hypothalamic circuit. *Tsukuba Conference 2021* (September 2021). Tsukuba, Japan (online).

Prokofeva K, Saito YC, Hirano A, Sakurai T. Mapping of sleep/wake-governing circuits of orexin-producing neurons. *The 44th Annual Meeting of the Japan Neuroscience Society* (July 2021). Kobe, Japan.

Selected Poster Presentations

Prokofeva K, Shibamiya M, Kawata R, Yoshihara C, Kuroda KO. Dissecting Shared Neural Mechanisms of Infant Care and Helping-like Behavior under Adversity in Female Mice. *The 49th Annual Meeting of the Japan Neuroscience Society* (July 2026). Kobe, Japan.

Prokofeva K, Shibamiya M, Kawata R, Yoshihara C, Kuroda KO. Neural Correlates and Neural Mechanisms of Cost-overcoming Helping-like Behavior in Female Mice. *International Symposium for the 20th Anniversary of the Japanese Society for Behavioral Neuroendocrinology* (July 2026). Tsukuba, Japan.

Prokofeva K, Shibamiya M, Kawata R, Yoshihara C, Kuroda KO. Exploring shared neuronal mechanisms of infant care and helping-like behavior in female mice. *The 4th Joint Poster Presentation for Young Scientists at RIKEN CBS* (March 2026). Tokyo, Japan.

Prokofeva K, Saito YC, Niwa Y, Mizuno S, Takahashi S, Hirano A, Sakurai T. A whole-brain input landscape of the GABA- and galaninergic neurons in the ventrolateral preoptic nucleus projecting to the lateral hypothalamus. *Society for Neuroscience Annual Meeting* (November 2022). San Diego, USA.

Academic & Professional Service

- Participate in undergraduate and Master's program entrance examinations, Institute of Science Tokyo.
- Coordinate laboratory-level processing of international student applications, including communication with prospective applicants and liaison with university administrative offices.
- Reviewer for *Scientific Reports*, *Current Biology*, and *Nature Neuroscience*.
- Session Chair, 49th Annual Meeting of the Japan Neuroscience Society, 2026.

Outreach & Mentoring

- The Women in Science Japan (WISJ) Mentorship Program 2026. Mentor for two PhD students at Japanese universities.
- Mentor, WISJ High School Outreach Initiative 2025.
- Organizing committee member, WISJ Flagship Event 2025.

Professional Memberships

Japan Neuroscience Society, Japanese Society for Behavioral Neuroendocrinology

Additional Professional Experience

2017 – 2018	Nurse Children's City Clinical Hospital No. 1 , Novosibirsk, Russia Pediatric Intensive Care Unit Part-time clinical employment during medical training
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Languages

Russian (native), English, Japanese